

Solutions to MP 204 Problem set 5

(1)

P. 1

This is quite similar to what we did when we showed that $\vec{E}$ & $\vec{B}$ satisfy the wave eq'n in vacuum, except (i) we're finding eq's for Φ and $\vec{A}$, and (ii) we're not assuming $\vec{J}$ and ρ vanish. Still, the eq's we get are extremely useful, and if you take MP465 at some point, you'll see them in action...

$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ and $\vec{\nabla} \cdot \vec{B} = 0$ are automatically satisfied if

$\vec{E} = -\vec{\nabla}\Phi - \frac{\partial \vec{A}}{\partial t}$ and $\vec{B} = \vec{\nabla} \times \vec{A}$, so it's the other two eq's that are interesting. Let's put $\vec{E}$ into Maxwell's First eq'n and see what happens:

$$\vec{\nabla} \cdot \vec{E} = \vec{\nabla} \cdot \left(-\vec{\nabla}\Phi - \frac{\partial \vec{A}}{\partial t} \right) = -\nabla^2 \Phi - \vec{\nabla} \cdot \frac{\partial \vec{A}}{\partial t} = \rho/\epsilon_0.$$

However, we can always compute partial derivatives in any order we like, so $\vec{\nabla} \cdot \frac{\partial \vec{A}}{\partial t} = \frac{\partial}{\partial t} (\vec{\nabla} \cdot \vec{A})$. However, we are assuming that the Coulomb condition $\vec{\nabla} \cdot \vec{A} = 0$ holds, so this term doesn't contribute at all. Thus, we find

$$\boxed{\nabla^2 \Phi = -\rho/\epsilon_0}$$

which is a well-known differential eq'n, Poisson's eq'n. If we can solve that, then we have Φ . In fact, as you'll see in MP465, the sol'n is

$$\Phi(t, \vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(t, \vec{r}') dV'}{|\vec{r} - \vec{r}'|}$$

so if we know ρ as a function of t, x, y and z , we can (in principle) find the scalar potential. (If you don't see where this comes from, don't worry - you probably haven't learned it yet.)

What about $\vec{A}$? Well, let's put $\vec{E}$ and $\vec{B}$ into Maxwell's Fourth eq'n:

$$\begin{aligned} \vec{\nabla} \times \vec{B} &= \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \mu_0 \vec{J} \\ \Rightarrow \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) &= \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial}{\partial t} \left(-\vec{\nabla}\Phi - \frac{\partial \vec{A}}{\partial t} \right) \\ \Rightarrow \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} &= \mu_0 \vec{J} - \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\vec{\nabla}\Phi) - \mu_0 \epsilon_0 \frac{\partial^2 \vec{A}}{\partial t^2}. \end{aligned}$$

But $\vec{\nabla} \cdot \vec{A} = 0$ is assumed, and $\frac{\partial}{\partial t} \vec{\nabla}\Phi = \vec{\nabla} \left(\frac{\partial \Phi}{\partial t} \right)$, so we get, with a little rearranging,

$$\boxed{\nabla^2 \vec{A} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{A}}{\partial t^2} = \mu_0 \epsilon_0 \vec{\nabla} \left(\frac{\partial \Phi}{\partial t} \right) - \mu_0 \vec{J}}$$

Thus, if we have $\Phi(t, \vec{r})$ from above and have $\vec{J}(t, \vec{r})$, then, in principle,

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we can find $\vec{A}(t, \vec{r})$. And once we have both Φ and $\vec{A}$, $\vec{E}$ and $\vec{B}$ follow immediately.

P.2

This is just putting the potentials into the formulas for $\vec{E}$ and $\vec{B}$ and checking the Lorenz condition:

(a) $\vec{E}$ is given by

$$\begin{aligned}\vec{E} &= -\vec{\nabla}\Phi - \frac{\partial \vec{A}}{\partial t} = -\left(\frac{\partial \Phi}{\partial x} + \frac{\partial A_x}{\partial t}\right)\hat{i} - \left(\frac{\partial \Phi}{\partial y} + \frac{\partial A_y}{\partial t}\right)\hat{j} - \left(\frac{\partial \Phi}{\partial z} + \frac{\partial A_z}{\partial t}\right)\hat{k} \\ &= -(0+0)\hat{i} - (0+0)\hat{j} - (0+B_0 y \cos \omega t)\hat{k} \\ &= \boxed{-B_0 \omega y \cos \omega t \hat{k}}.\end{aligned}$$

 $\vec{B}$ is

$$\begin{aligned}\vec{B} &= \vec{\nabla} \times \vec{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{i} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{j} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{k} \\ &= (B_0 \sin \omega t - 0)\hat{i} + (0-0)\hat{j} + (0-0)\hat{k} \\ &= \boxed{B_0 \sin \omega t \hat{i}}\end{aligned}$$

(b) Now we check the Lorenz condition:

$$\begin{aligned}\mu_0 \epsilon_0 \frac{\partial \Phi}{\partial t} + \vec{\nabla} \cdot \vec{A} &= \mu_0 \epsilon_0 \frac{\partial}{\partial t} \left(\frac{2B_0}{\mu_0 \epsilon_0} t \right) + \frac{\partial}{\partial x} (B_0 x) + \frac{\partial}{\partial y} (0) + \frac{\partial}{\partial z} (B_0 (y \sin \omega t - 3z)) \\ &= \mu_0 \epsilon_0 \left(\frac{2B_0}{\mu_0 \epsilon_0} \right) + B_0 - 3B_0 = \boxed{0}\end{aligned}$$

so these potentials do indeed satisfy the Lorenz condition.

P.3

Now we do similar calculations with slightly different potentials:

$$\begin{aligned}(a) \vec{E} &= -\left(\frac{\partial \Phi}{\partial x} + \frac{\partial A_x}{\partial t}\right)\hat{i} - \left(\frac{\partial \Phi}{\partial y} + \frac{\partial A_y}{\partial t}\right)\hat{j} - \left(\frac{\partial \Phi}{\partial z} + \frac{\partial A_z}{\partial t}\right)\hat{k} \\ &= -(0+0)\hat{i} - (0+0)\hat{j} - (0 + \omega B_0 y \sin \omega t)\hat{k} \\ &= \boxed{-B_0 \omega y \sin \omega t \hat{k}}\end{aligned}$$

$$\begin{aligned}\vec{B} &= \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{i} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{j} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{k} \\ &= (B_0 \sin \omega t - 0)\hat{i} + (0-0)\hat{j} + (0-0)\hat{k}\end{aligned}$$

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$$= B_0 \sin \omega t \hat{i}$$

So, yes, $\vec{E}$ and $\vec{B}$ are the same as in P2, even though Φ and $\vec{A}$ are not. This illustrates the fact that the potentials are not unique: there are a huge number which give the same electric and magnetic fields.

(b) Let's check the Lorenz cond'n:

$$\begin{aligned} \mu_0 \epsilon_0 \frac{\partial \Phi}{\partial t} + \vec{\nabla} \cdot \vec{A} &= \mu_0 \epsilon_0 \frac{\partial}{\partial t} \left(-\frac{3B_0}{\mu_0 \epsilon_0} t \right) + \frac{\partial}{\partial x} (2B_0 x) + \frac{\partial}{\partial y} (-2B_0 y) + \frac{\partial}{\partial z} (B_0 y \sin \omega t) \\ &= \mu_0 \epsilon_0 \left(-\frac{3B_0}{\mu_0 \epsilon_0} \right) + (2B_0) + (-2B_0) + 0 \\ &= -3B_0 \end{aligned}$$

which is not zero. Thus Φ and $\vec{A}$ do not satisfy the Lorenz cond'n. However, they do satisfy the Coulomb cond'n.

$$\begin{aligned} \vec{\nabla} \cdot \vec{A} &= \frac{\partial}{\partial x} (2B_0 x) + \frac{\partial}{\partial y} (-2B_0 y) + \frac{\partial}{\partial z} (B_0 y \sin \omega t) \\ &= 2B_0 - 2B_0 = 0. \end{aligned}$$

So we can always pick Φ & $\vec{A}$ that (i) give us the $\vec{E}$ and $\vec{B}$ we want and (ii) satisfy any number of cond'ns we might like to require of them. This can be very handy sometimes.